

VALPROIC ACID and ADVERSE EFFECTS

- FDA approved uses
 - Epilepsy
 - Migraine prophylaxis
 - Mania, bipolar
- Contraindications
 - Hepatic dz, urea cycle disorders, Concurrent use with Lamotrigine (glucuronidation)
- Formulations
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Depakote (Divalproex sodium)DR (divided dose), ER (1x/day), need slightly higher dose for equivalence	Delayed release coating, no crush/chew
Depakene (Valproic acid),	oral tablet and liquid (DON'T USE→ GI AE)
Depakote Sprinkles	only for epilepsy, one dose

- Adverse Effects:

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Hair loss	Acute Liver Failure
Weight gain	DRESS syndrome
Sedation	Neural tube defects
N/V/Dizzy/dyspepsia	Hyperammonemia
Pancreatitis	GI

THE BIG 3

- Typically first month, more commonly 1-2 weeks
- Elevated LFT, Elevated ammonia, abd pain, lethargic, fever, rash, Eosinophilia, elevated WBC, low platelets

- Tx

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N-Acetylcystein (NAC) 150mg/kg/hr over one hour, followed by 12.5mg/kg/hr for 4 hrs, then continuous infusion 6.25mg/kg/hr for 67hrs	L-Carnitine For symptomatic patients with hyperammonemia: Loading dose of 100 mg/kg intravenously (6 g maximum) over 30 minutes followed by maintenance doses of 15 mg/kg every 4 hours over 10-30 minutes. For asymptomatic patients with hyperammonemia: Oral doses of 100 mg/kg/day divided every 6 hours (3 g/day maximum).	Lactulose
Prednisone	Famotidine	Diphenhydramine

- Obviously D/C VA, but symptoms may progress despite discontinuing VA
- Liver transplant
- General Monitoring
 - Before: CBC, BMP, LFT
 - W/in 1wk: same, trough level
 - 6mo, Annually